

Souhaieb Elamri

Electrical Engineering Student | Computer Vision and Industrial AI for Automation

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PROFILE

Electrical Engineering student at ENSIT with a Master's in Advanced Robotics and Artificial Intelligence. Substantial hands-on experience in computer vision, with strong technical skills in embedded systems, industrial automation, and Industrial AI. Seeking a final-year engineering internship starting February 2027.

PROFESSIONAL EXPERIENCE

- Summer Intern – WIC MIC GROUP**06/2026 – 08/2026 | Bizerte, Tunisia
- Tools: Python, YOLO11, YOLO11-Pose, DocTR, SVM, CatVTON, OpenCV, FastAPI, Streamlit, WebRTC, Docker, SQL Server
- Automated production ticket digitization using YOLO11, SVM, and FastAPI, completely eliminating manual data entry.
 - Built a real-time virtual garment try-on platform (CatVTON, WebRTC) utilizing ArUco markers for centimeter-level spatial accuracy.
- Master's Thesis Intern – WIC MIC GROUP**02/2025 – 05/2025 | Bizerte, Tunisia
- Tools: Python, OpenCV, PyTorch, YOLO11-Pose, CUDA, Docker, SQL Server
- Engineered a real-time AI garment inspection system, managing everything from massive data annotation to OpenCV multi-camera calibration, achieving ±0.3 cm precision.
- Bachelor's Project Intern – HYCO Industries – Metal Construction**02/2023 – 05/2023 | Nabeul, Tunisia
- Retrofitted a manual metal fabrication machine into an automated multi-axis CNC system, integrating stepper motor drivers and configuring Mach3 control software via a Raspberry Pi interface.
- Technical Intern – CMR Tunisia Ship Repairs**01/2023 – 02/2023 | Bizerte, Tunisia
- Automated a marine basin control system by developing a PLC/HMI architecture using Siemens PLCs, Ladder Logic, and TIA Portal for real-time pump, valve, and water-level control.

EDUCATION

- National Engineering Diploma in Electrical Engineering**2025 – 2027 (expected)
National Higher School of Engineers of Tunis (ENSIT)
- Master's Degree in Advanced Robotics and Artificial Intelligence**2023 – 2025
Higher Institute of Technological Studies of Bizerte
- Bachelor's Degree in Electrical Engineering – Industrial Automation**2020 – 2023
Higher Institute of Technological Studies of Nabeul

ACADEMIC PROJECTS

- Smart Embedded System for Wildfire Prevention**2026 – Present
- Designed an embedded environmental monitoring node using a Raspberry Pi. Integrated SPI/I2C sensors to process 4 environmental metrics (CO2, moisture, temperature, wind) and transmit low-latency hazard alerts.
- Intelligent Mobile Robot**2025 – 2026
- Designed a mobile robotics platform using ESP8266 and Node-RED. Engineered PWM motor controls and programmed I2C/ADC sensor data acquisition to provide real-time IoT telemetry.

TECHNICAL SKILLS

- Embedded & Industrial:** Raspberry Pi, Arduino, ESP32, STM32, FPGA, VHDL, TIA Portal, PLC, CNC, LabVIEW
- AI & Computer Vision:** TensorFlow, PyTorch, OpenCV, MediaPipe, YOLO, Object Detection, Pose Estimation, OCR
- Tools & Languages:** Python, C, MATLAB/Simulink, SolidWorks, FastAPI, SQL Server, Docker, Git

CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals (AI-900)** ☑ – 05/2024
- Microsoft Certified: Security, Compliance, and Identity Fundamentals (SC-900)** ☑ – 10/2024
- Six Sigma Yellow Belt** ☑ – 01/2025

LANGUAGES

- French** – Fluent | **English** – Fluent | **Arabic** – Native